

The Coordinate System as Ontological Foundation: Why Existence Requires a Frame Before It Can Change

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ABSTRACT

This paper proposes a philosophical-mathematical thesis: that existence, conceived as universal and continuous change, necessarily presupposes a stable ontological substrate which we term the Coordinate System (Ma'lam). The central argument proceeds in three stages. First, any change — whether physical, temporal, or structural — is formally representable as a function. Second, every function requires a frame of definition: without coordinates, a function cannot be plotted, measured, or said to exist. Third, the Coordinate System is itself infinite and unchanging, constituting the absolute precondition for all becoming. A key consequence follows: chance may explain the motion of functions within the frame, but it cannot explain the existence of the frame itself. We further argue that physical laws tend toward simplicity at scale — from quantum mechanics through Newtonian mechanics to relativistic unification — suggesting that the equation of existence may be of deceptive simplicity. The Collatz function ($3x + 1$), independently derived by the author from Modulo-9 wave mechanics, is offered as a structural analogue. Finally, we argue that the void (al-'adam) is ontologically impossible: the Coordinate System is everywhere, and the apparent void is merely a region in which no function has yet been inscribed.

Keywords: Ontology of existence, coordinate system, function, change, void, Collatz conjecture, Modulo-9, philosophy of mathematics, cosmological simplicity.

1. Introduction: The Problem of Existence and Change

Philosophy has long grappled with a foundational question: what is existence, at its most irreducible level? The present paper approaches this question not through metaphysical speculation alone, but through the structural language of mathematics — specifically, through the concept of the function and its necessary substrate: the coordinate system.

The guiding intuition is precise: **everything that exists, changes**. Matter moves, energy transforms, time flows, structures emerge and dissolve. Change is not incidental to existence — it appears to be its defining characteristic. If this is granted, a natural mathematical translation follows: any aspect of the universe is formally representable as a function. A function is precisely a rule of change: it maps states to states, moments to moments.

But a function cannot exist in isolation. It requires a domain, a range, and — crucially — a frame in which to be plotted and measured. This frame is the coordinate system. Without it, the function has no location, no graph, no meaning. The coordinate system is not itself a function: it does not change. It is the silent, infinite, stable precondition for all change to occur.

This paper argues that this mathematical observation has deep ontological implications. The Coordinate System is not a human construct but the fundamental ontological substrate of existence — what philosophers might call the Absolute, and what mathematics calls the infinite space of all possible definitions.

2. Formal Framework: Existence, Functions, and the Coordinate System

2.1 Existence as Universal Change

We begin with an axiomatic observation grounded in both physics and phenomenology:

Axiom 1 (Universal Change): Every observable entity undergoes change. No state of the universe is identical to a preceding state.

This is consistent with thermodynamics (entropy increases monotonically), quantum mechanics (wave functions evolve continuously), and general relativity (spacetime geometry is dynamically determined by matter-energy content). Change is the universal condition of existence itself.

2.2 Change as Function

If existence is change, and change is the mapping of states across time, then existence is formally representable as a function. Let:

$$f: S \times T \rightarrow S$$

where S is the space of possible states and T is the temporal parameter. This function describes how any configuration evolves. More generally, any observable phenomenon — physical, biological, cognitive — can be represented as a function from an initial condition to a subsequent state.

2.3 The Necessity of the Coordinate System

A function requires: (i) a domain — the set of valid inputs; (ii) a codomain — the set of possible outputs; (iii) a frame of reference — the coordinate system in which the function can be plotted, measured, and said to exist.

Without the coordinate system, a function is not merely unmeasured — it is undefined. We therefore define:

Definition (Coordinate System / Ma'lam): The Coordinate System is the infinite, stable, and unchanging ontological substrate that constitutes the necessary precondition for any function — and therefore any change, and therefore any existence — to be defined.

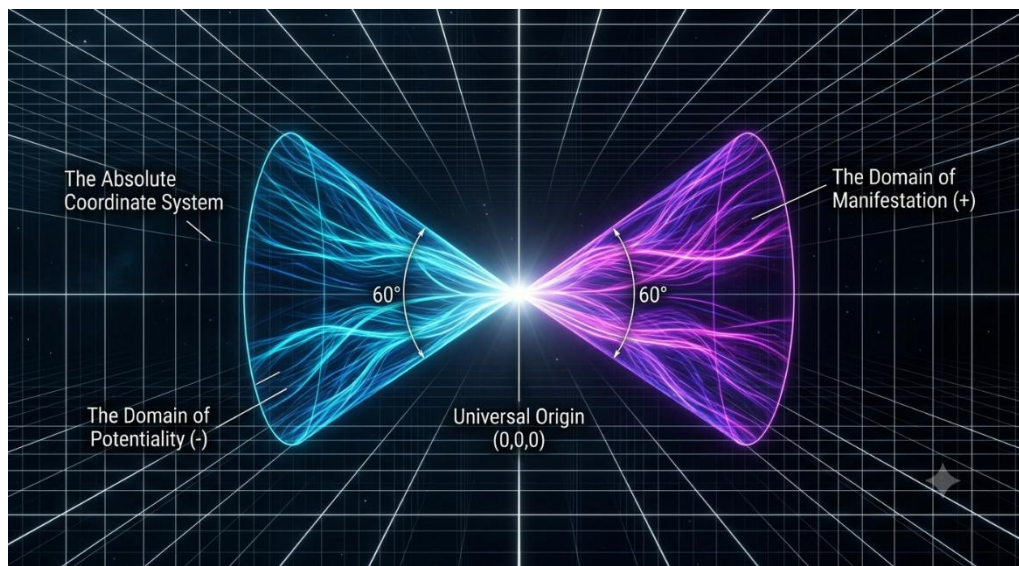


Figure 1. The Dual-Cone Model of Ontological Substrate. The image illustrates the Coordinate System (Ma'lam) as an infinite, unchanging 3D grid that precedes all existence. The symmetrical double-cones (60° opening) represent the expansion of functions and change from the singular Universal Origin (0,0,0). The grid's presence in the dark regions—where no functions are yet inscribed—visually demonstrates the impossibility of 'the void' (al-'adam), confirming that the frame of possibility is everywhere and always present.

The Coordinate System does not itself change. It is not a function. It is the frame in which all functions are inscribed.

2.4 The Asymmetry Between Frame and Function

Proposition 1: Chance may, in principle, account for the specific trajectory of a function within the Coordinate System. It cannot account for the existence of the Coordinate System itself.

To see this: suppose we concede that the universe's initial conditions arose by chance. This concession still presupposes a space — a frame of logical and mathematical possibility — in which chance could operate. The roll of a die presupposes the existence of the die, the table, and the physical laws that govern the roll. Chance is always chance *within* a structure. The structure cannot itself be the product of chance without infinite regress.

The Coordinate System is precisely this irreducible structure. It is not contingent. It is the precondition of contingency itself.

3. The Impossibility of the Void

The apparent void is not the absence of the Coordinate System. It is the absence of functions inscribed within it. The grid exists everywhere — it is only that, in certain regions, no value has yet been plotted. The void is not nothingness: it is the Coordinate System in an unoccupied state.

What we call 'empty space' in physics — the quantum vacuum, the interstellar void — is not ontological emptiness. It is a region of the Coordinate System in which fields fluctuate, potentials are defined, and geometry is curved. The frame is always present.

Theorem 1 (Impossibility of the Void): True ontological nothingness — the absence of the Coordinate System — is impossible. The apparent void is the Coordinate System in the absence of inscribed functions. The Coordinate System itself is everywhere and always.

This resolves, or rather dissolves, the classical question 'why is there something rather than nothing?' The question is malformed: it assumes that nothingness is a coherent ontological possibility. On the present account, the Coordinate System is not 'something' in the ordinary sense — it is the frame of possibility itself, which cannot coherently be said to be absent.

4. The Cosmological Tendency Toward Simplicity

A remarkable feature of physical knowledge is its behavior under change of scale. As we move from the microscopic to the macroscopic, the equations governing reality tend to become simpler, not more complex.

At the quantum scale, the universe is governed by the Schrödinger equation and quantum field theory — dense, non-intuitive mathematics that resists classical visualization. At the intermediate scale, Newton's three laws emerge as an extraordinarily accurate description of motion, expressible in a single

line. At the cosmological scale, Einstein's field equations unify gravity, geometry, and energy into a single structural relationship.

The pattern is striking: *the larger the scale at which we view the universe, the simpler the governing equation becomes*. Complexity emerges at the local level; simplicity is a property of the global frame.

Hypothesis (Ontological Simplicity): If there exists a fundamental equation of existence, it is likely to be of deceptive simplicity: a rule that generates irreducible complexity from a minimal prescription.

This is not a proof. It is a structural tendency, observable across the history of physics, that points toward a deep feature of the Coordinate System: the frame is simple; the functions inscribed within it generate complexity.

5. The Collatz Function as Structural Analogue

In prior work (Charrat, 2025), the author independently derived the Collatz linear form $3x + 1$ from the internal wave mechanics of the Modulo-9 number system. The derivation proceeded without assuming the Collatz conjecture. The function emerged as a necessary identity from the structural balancing of modular wave equations — from the nine-fold accumulation of wave variables and the invariant constant obtained after digital-root reduction.

We do not claim that $3x + 1$ is the equation of existence. We offer it as a *structural analogue* — a concrete mathematical object exhibiting the properties we have argued the equation of existence must possess:

(i) Simplicity of form: The rule is expressible in four characters. It requires no advanced mathematics to state.

(ii) Irreducible complexity of dynamics: Despite its simplicity, no proof exists that all trajectories converge. The function generates sequences of extraordinary length and apparent randomness.

(iii) Universal reach: The function is defined for all positive integers — it acts on the entire domain of its coordinate system.

(iv) Structural origin: The constants 3 and 1 are not arbitrary. They emerge from the internal topology of the Modulo-9 system — from the wave structure of the integers themselves.

$$C(n) = 3n + 1 \quad (\text{if } n \text{ is odd})$$

$$C(n) = n / 2 \quad (\text{if } n \text{ is even})$$

The question posed by the Collatz conjecture — do all trajectories return to 1? — mirrors, at a structural level, a deeper cosmological question: does all change, however complex, return to a fundamental unity? The Coordinate System is that unity: the frame to which every function, however elaborated, ultimately belongs.

6. Discussion: Epistemic Status and Philosophical Lineage

It is important to clarify the epistemic status of this framework. This is not a physical theory in the sense of general relativity or quantum mechanics. It does not generate predictions testable by experiment. It is a philosophical-mathematical framework — an attempt to identify the structural conditions that any theory of existence must satisfy.

The framework belongs to what might be called *mathematical ontology*: the use of formal mathematical structures to illuminate questions that classical philosophy has addressed through argument alone. In this tradition, we follow Leibniz (who saw the universe as the realization of the optimal among all possible worlds — a choice made within a coordinate system of possibilities), Spinoza (for whom the infinite substance was the ground of all finite modes), and more recently, Tegmark, who has argued that the universe is, at its deepest level, a mathematical structure.

What is original in the present work is the specific identification of the Coordinate System — the mathematical frame — as the ontological absolute, combined with: (i) the argument that the void is impossible because the frame is everywhere; (ii) the argument that chance presupposes structure and therefore cannot account for the frame; and (iii) the derivation of these conclusions from, and their connection to, a specific mathematical framework — the Numerical Origin Theory — developed independently by the author.

7. Conclusion

We have argued for the following connected theses:

1. Existence is universal change, and change is formally representable as a function.
2. Every function presupposes a Coordinate System — a stable, infinite, unchanging frame of definition.
3. The Coordinate System is the ontological absolute: the precondition of all existence, not itself subject to change or chance.
4. Chance may explain the trajectory of functions within the frame; it cannot explain the existence of the frame.

5. The apparent void is ontologically impossible: the Coordinate System is present everywhere.
6. Physical knowledge tends toward simplicity at scale, suggesting the fundamental equation of existence is simple in form and infinite in implication.
7. The Collatz function, derived from Modulo-9 wave mechanics, offers a structural analogue: minimal rule, infinite complexity, universal domain.

The deepest question in philosophy is not 'why is there something rather than nothing?' — that question assumes nothingness is coherent. The deeper question is: what is the structure that makes existence — change, function, becoming — possible at all? We have proposed that this structure is the Coordinate System: infinite, stable, and everywhere present. It is the frame before the first function was ever inscribed.

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